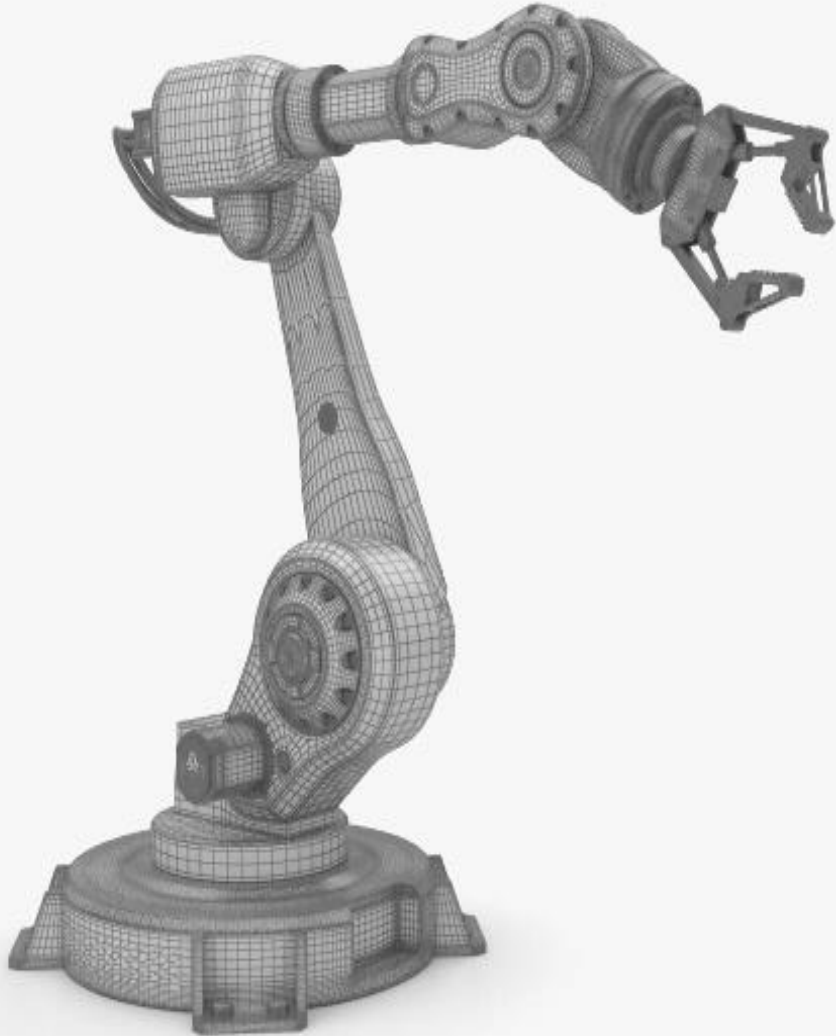




Robotic Systems

Chapter IV

Forward Kinematics

Prof. Dr. Ibrahim Hamarash
University of Kurdistan-Hewlêr

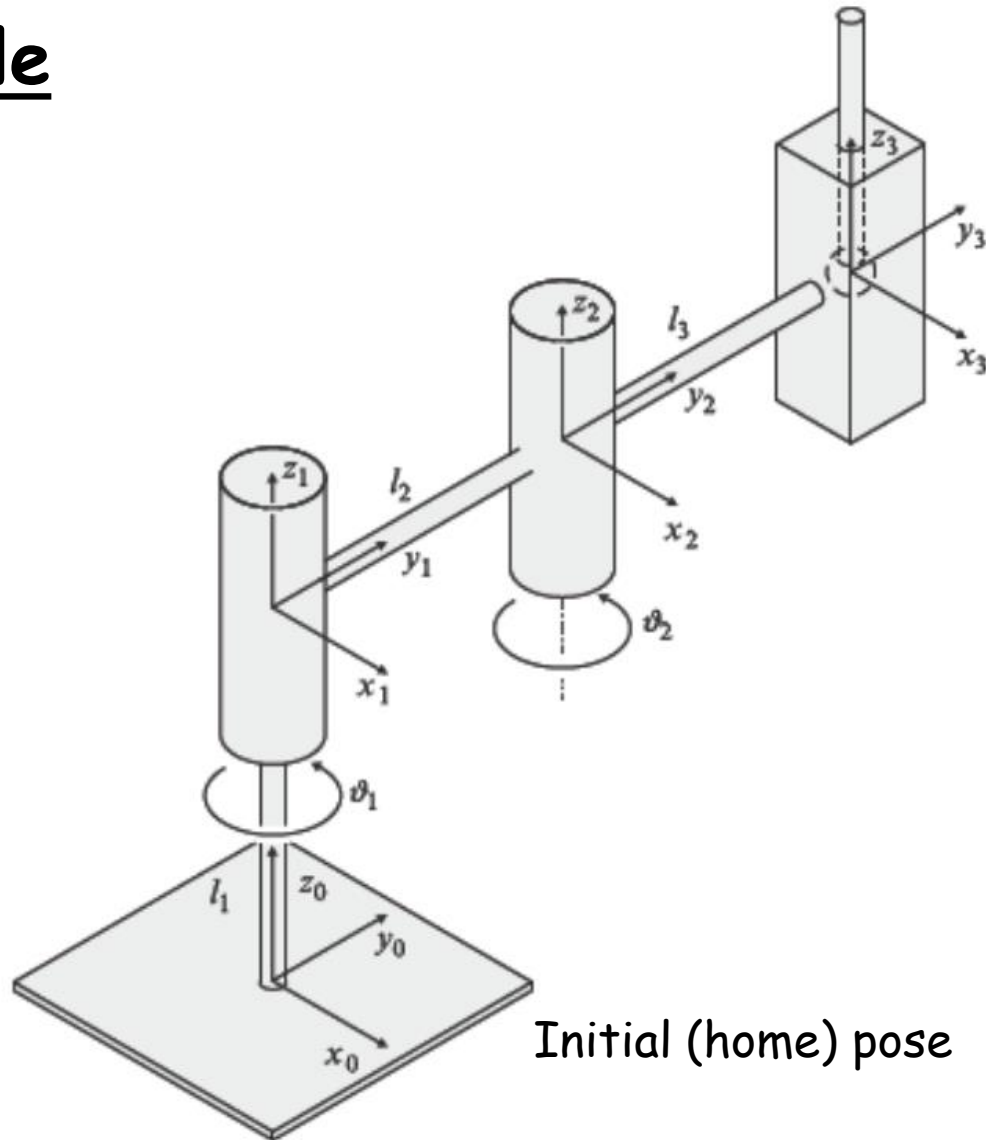
Manipulator Kinematics

The problem of manipulator kinematics is to describe the motion of a manipulator without consideration of the forces and torques causing the motion.

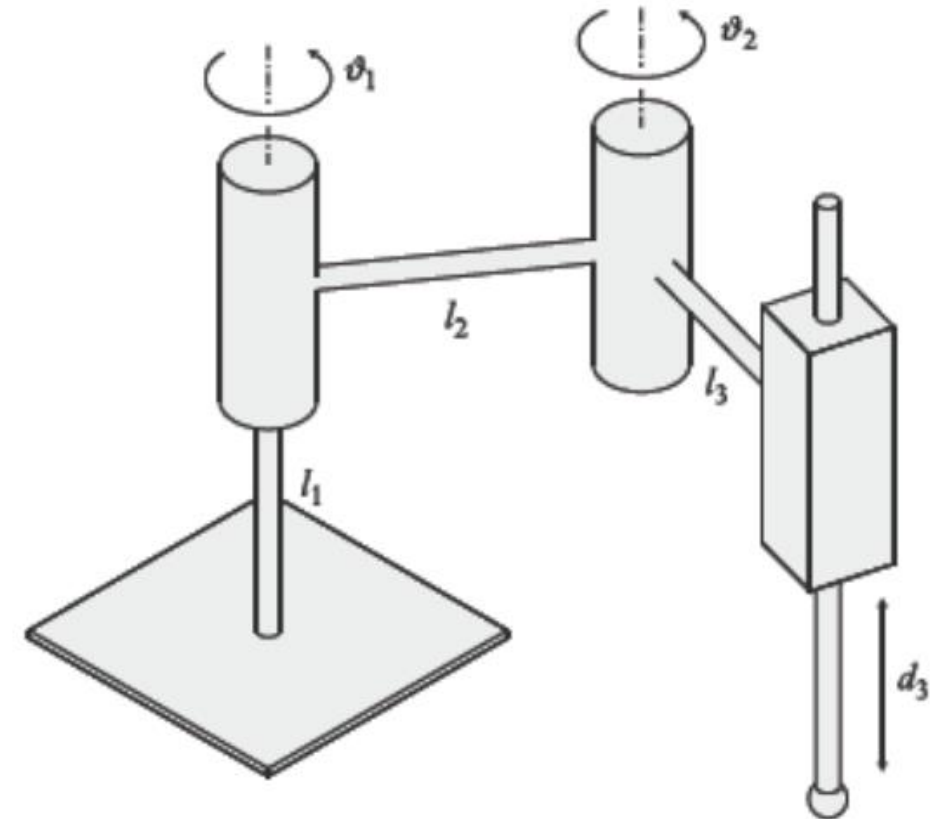
Forward Kinematics

The calculation of the position and orientation of the end-effector frame from joint coordinates.

Example



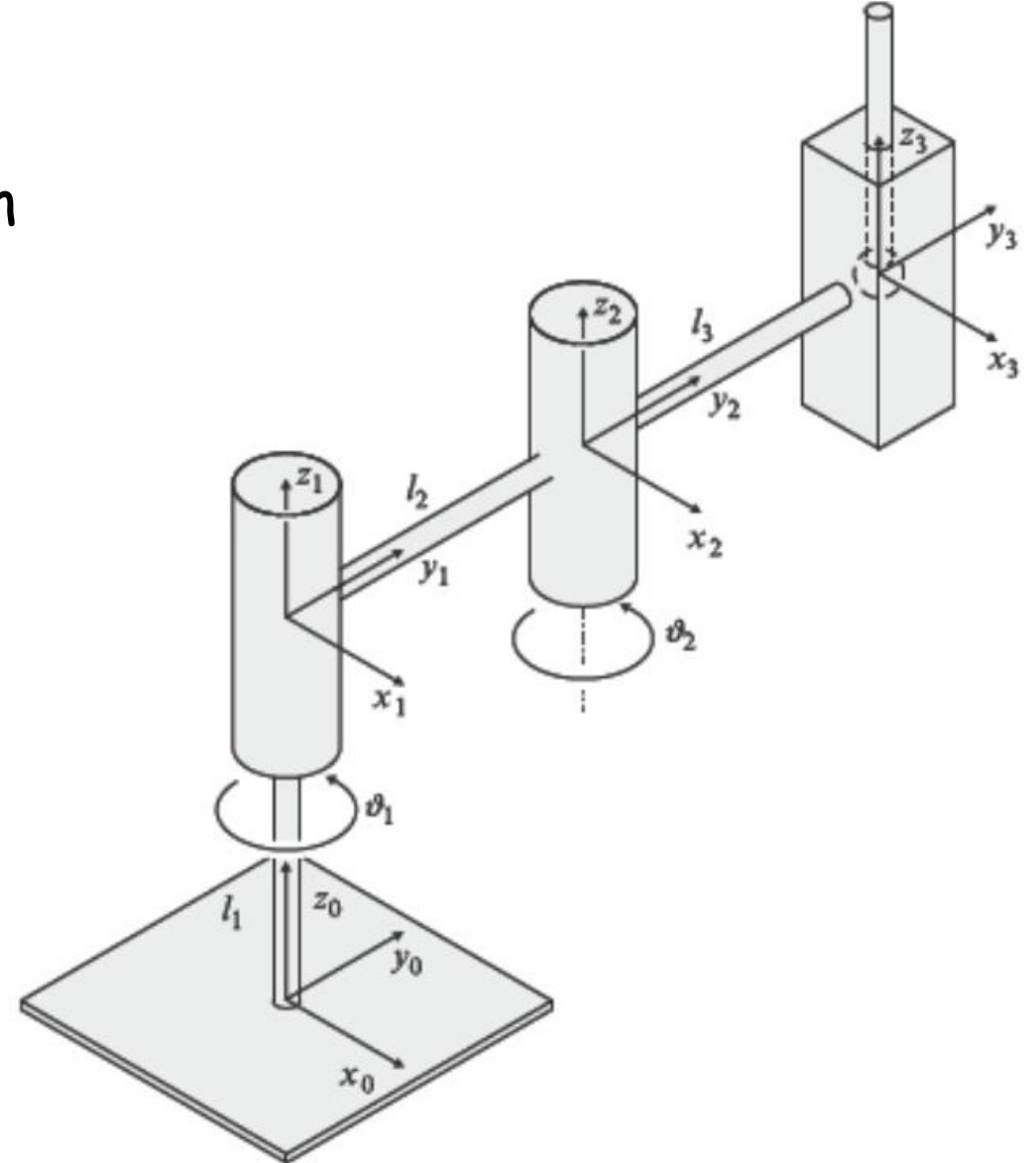
Initial (home) pose



Arbitrary pose

The pose of the end frame x_3, y_3, z_3 with respect to the reference frame x_0, y_0, z_0 is expressed by the following post multiplication of three pairs of homogenous transformation matrices:

$${}^0\mathbf{H}_3 = ({}^0\mathbf{H}_1\mathbf{D}_1) \cdot ({}^1\mathbf{H}_2\mathbf{D}_2) \cdot ({}^2\mathbf{H}_3\mathbf{D}_3).$$



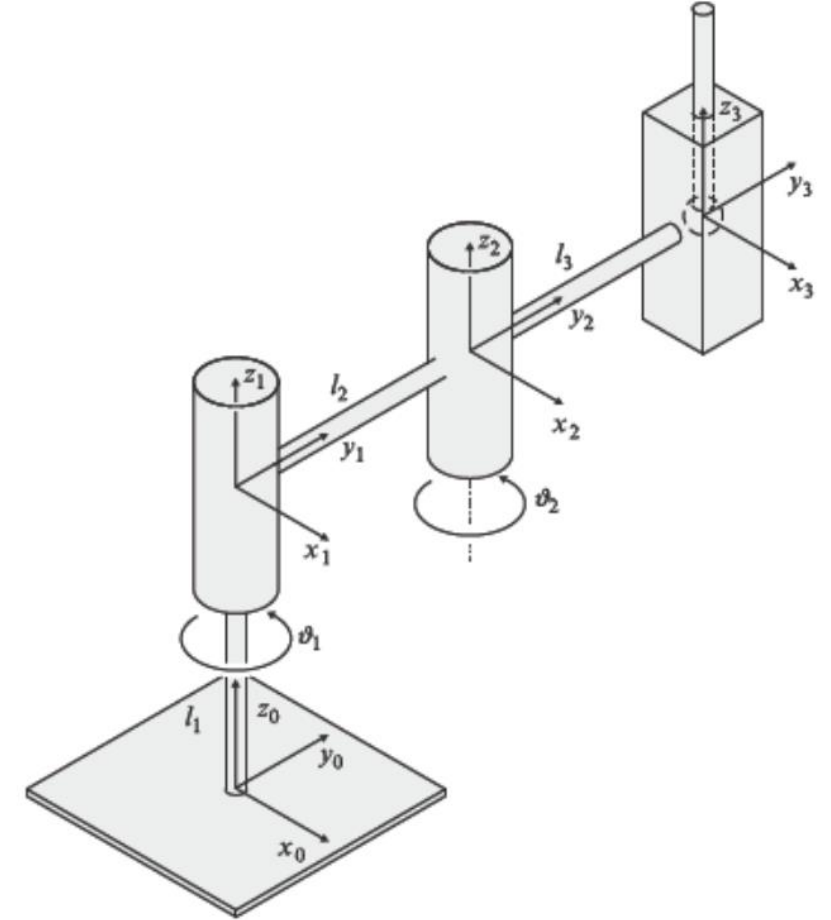
$${}^0\mathbf{H}_1\mathbf{D}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1\mathbf{H}_2\mathbf{D}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c2 & -s2 & 0 & 0 \\ s2 & c2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c2 & -s2 & 0 & 0 \\ s2 & c2 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2\mathbf{H}_3\mathbf{D}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_3 \\ 0 & 0 & 1 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

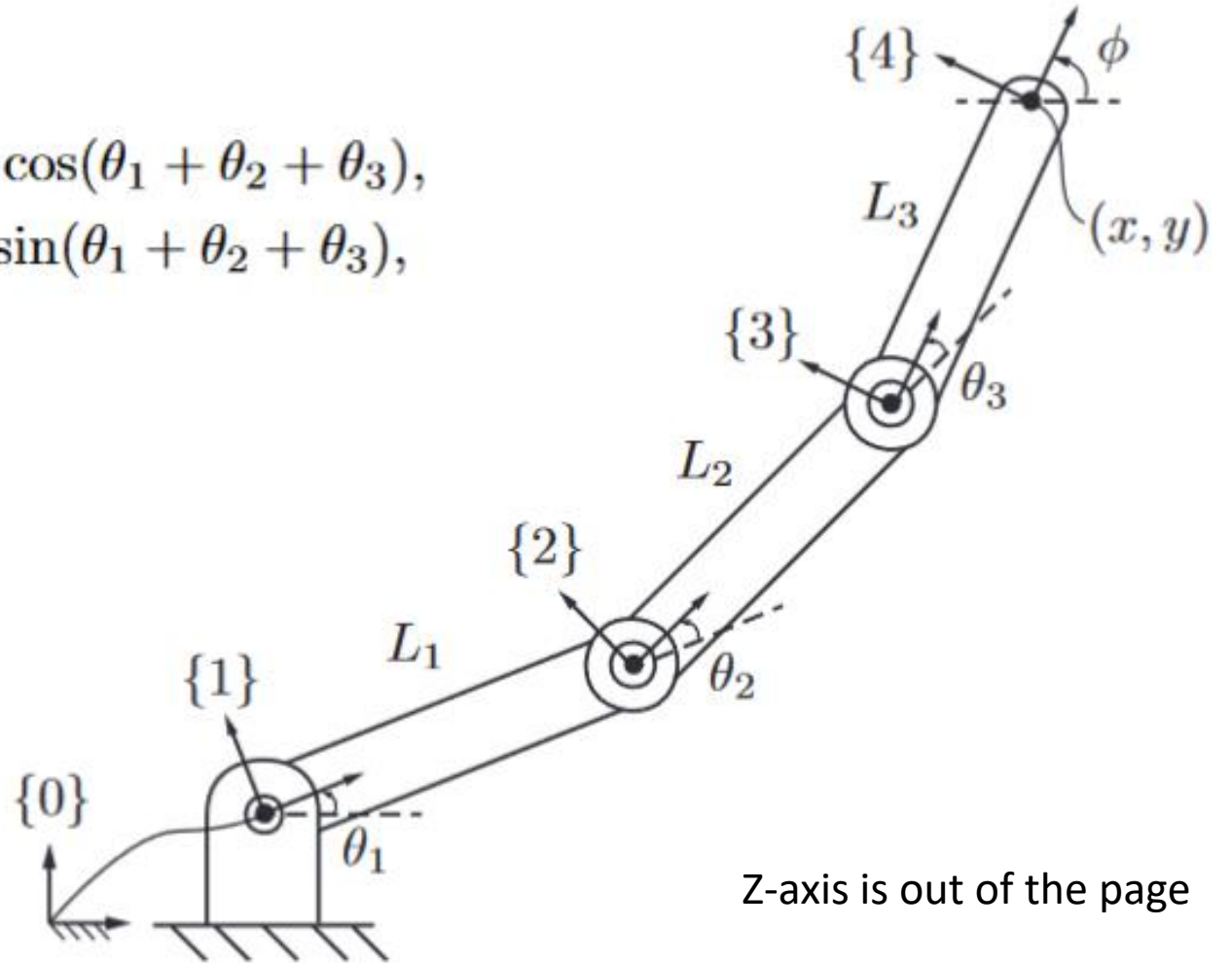
$${}^0\mathbf{H}_3 = \begin{bmatrix} c12 & -s12 & 0 & -l_3s12 - l_2s1 \\ s12 & c12 & 0 & l_3c12 + l_2c1 \\ 0 & 0 & 1 & l_1 - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$c12 = \cos(\vartheta_1 + \vartheta_2) = c1c2 - s1s2 \text{ and } s12 = \sin(\vartheta_1 + \vartheta_2) = s1c2 + c1s2.$$



Example: open chain

$$\begin{aligned} x &= L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3), \\ y &= L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3), \\ \phi &= \theta_1 + \theta_2 + \theta_3. \end{aligned}$$

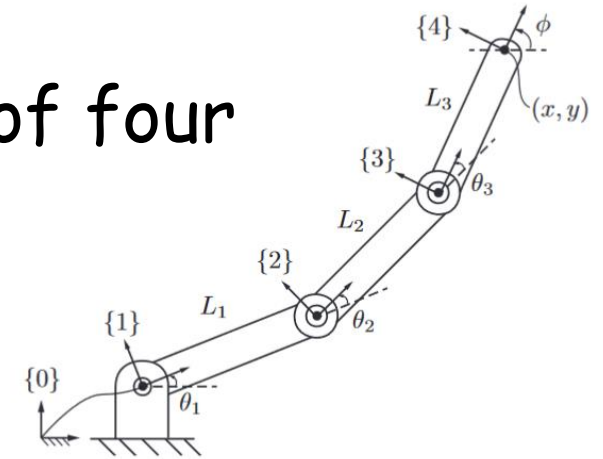


Z-axis is out of the page

Example

The forward kinematics can be written as a product of four homogeneous transformation matrices:

$$T_{04} = T_{01}T_{12}T_{23}T_{34},$$



$$T_{01} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{12} = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & L_1 \\ \sin \theta_2 & \cos \theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{23} = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & L_2 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{34} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

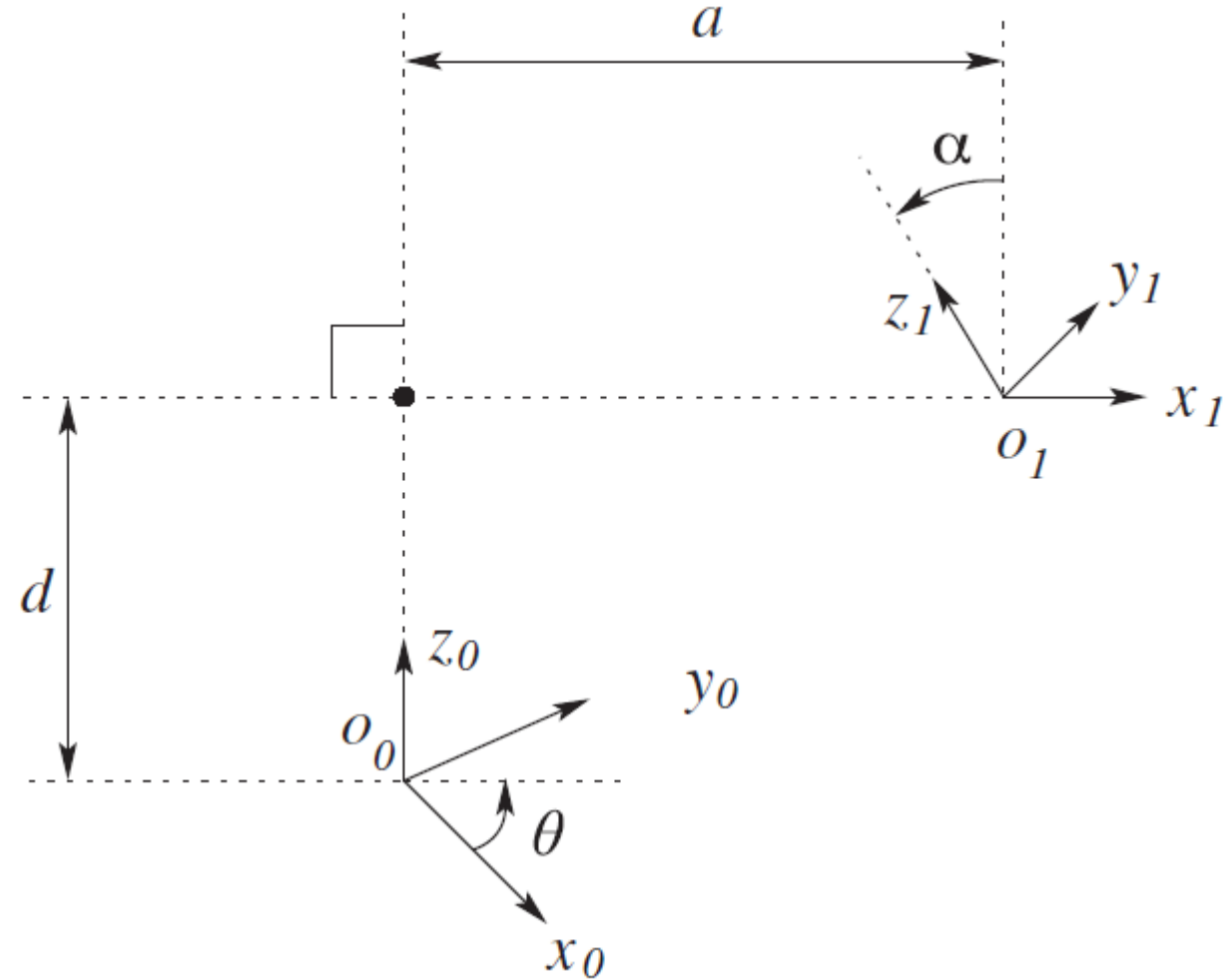
Transformation Example

$$\begin{aligned}
 A_i &= \text{Rot}_{z,\theta_i} \text{Trans}_{z,d_i} \text{Trans}_{x,a_i} \text{Rot}_{x,\alpha_i} \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &\quad \times \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_ic_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_is_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Note:

The four quantities θ_i , a_i , d_i , α_i are parameters associated with link i and joint i .

The four parameters a_i ; θ_i ; d_i , and α_i in are generally given the names **link length**, **link twist**, **link offset**, and **joint angle**, respectively.

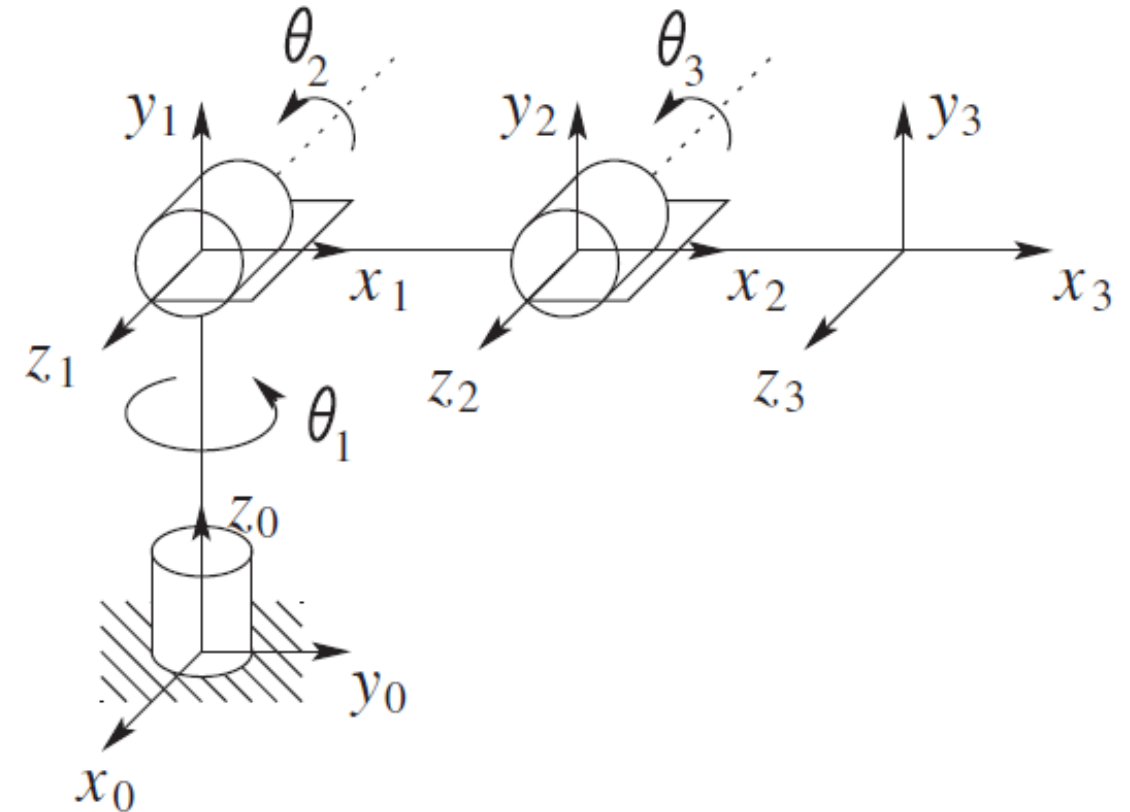


Coordinate Frame

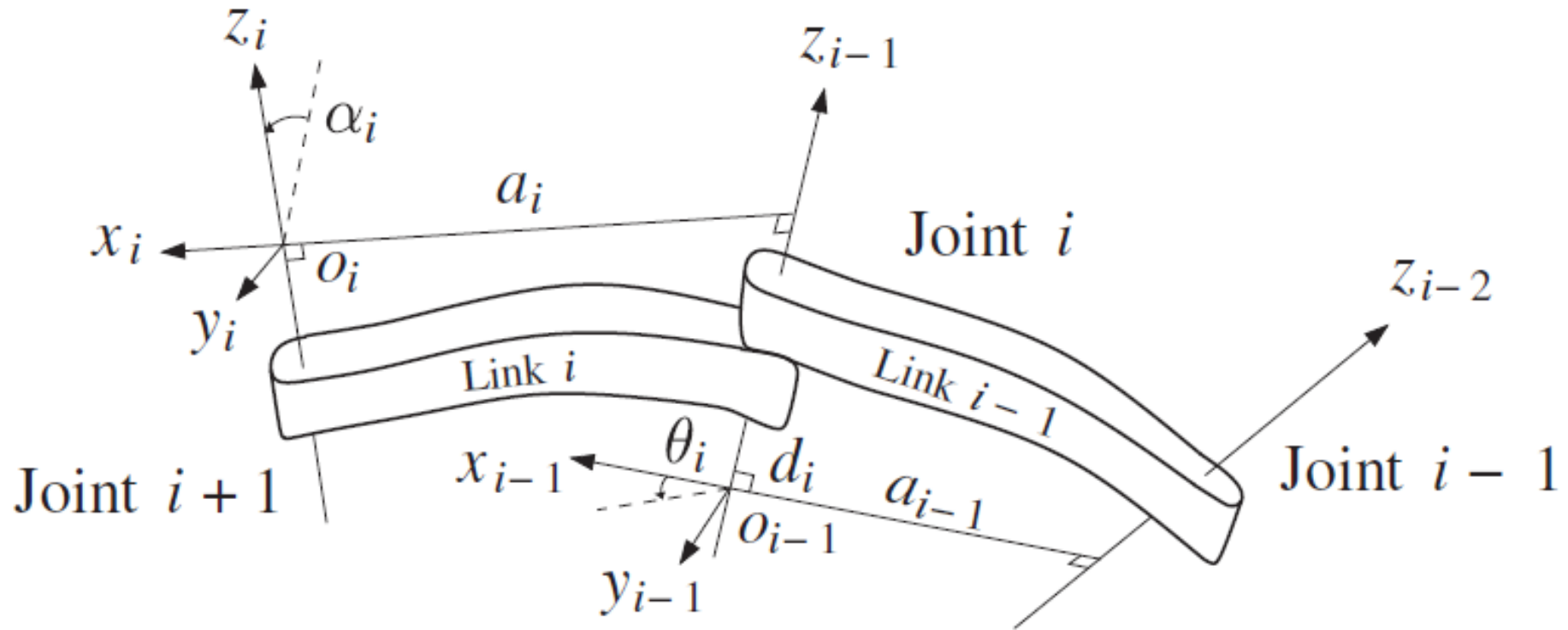
To perform the kinematic analysis, we attach a coordinate frame rigidly to each link.

In particular, we attach $o_i x_i y_i z_i$ to link i . This means that, whatever motion the robot executes, the coordinates of each point on link i are constant when expressed in the i -th coordinate frame.

Furthermore, when joint i is actuated, link i and its attached frame, $o_i x_i y_i z_i$, experience a resulting motion. The frame $o_0 x_0 y_0 z_0$, which is attached to the robot base, is referred to as the base frame, inertial frame or world frame.



Denavit-Hartenberg (DH) coordinate frame assignment

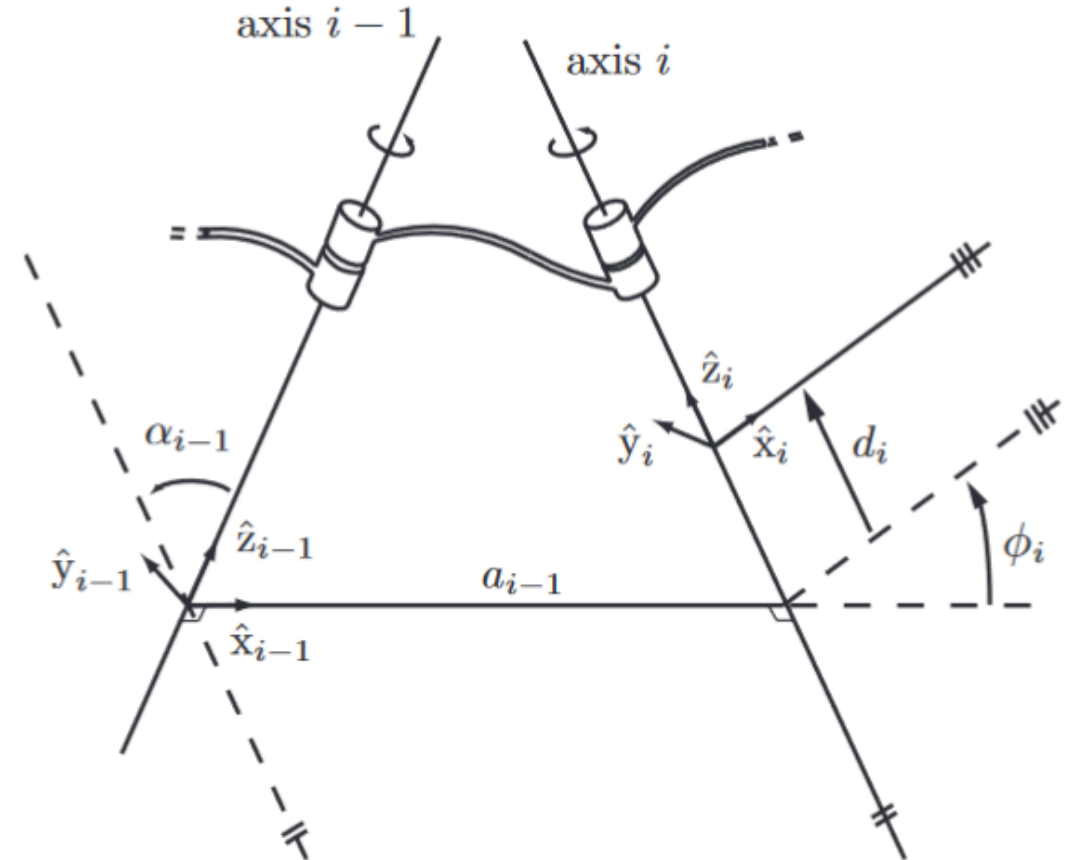


Denavit-Hartenberg Convention

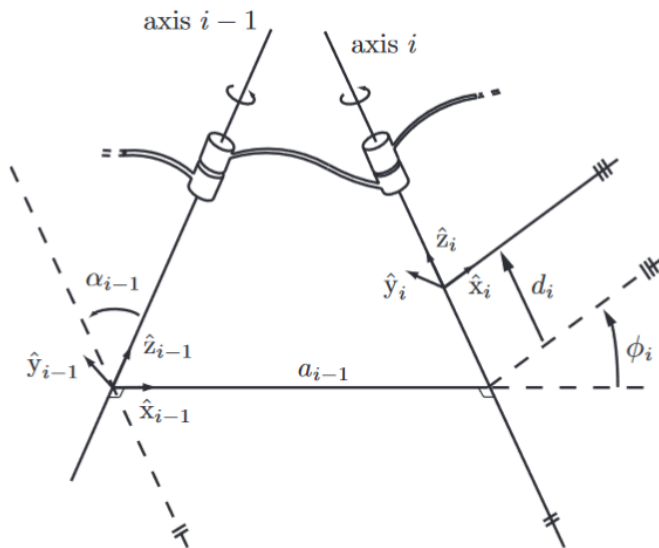
The forward kinematic is given by:

$$T_{0n}(\theta_1, \dots, \theta_n) = T_{01}(\theta_1)T_{12}(\theta_2) \cdots T_{n-1,n}(\theta_n)$$

With a set of rules for assigning link frames.

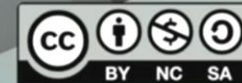
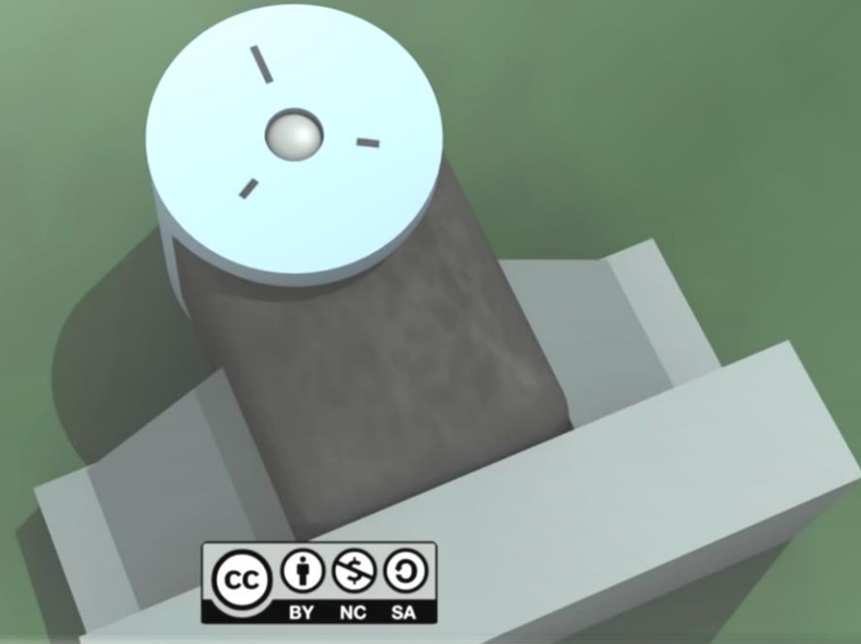


Video Segment



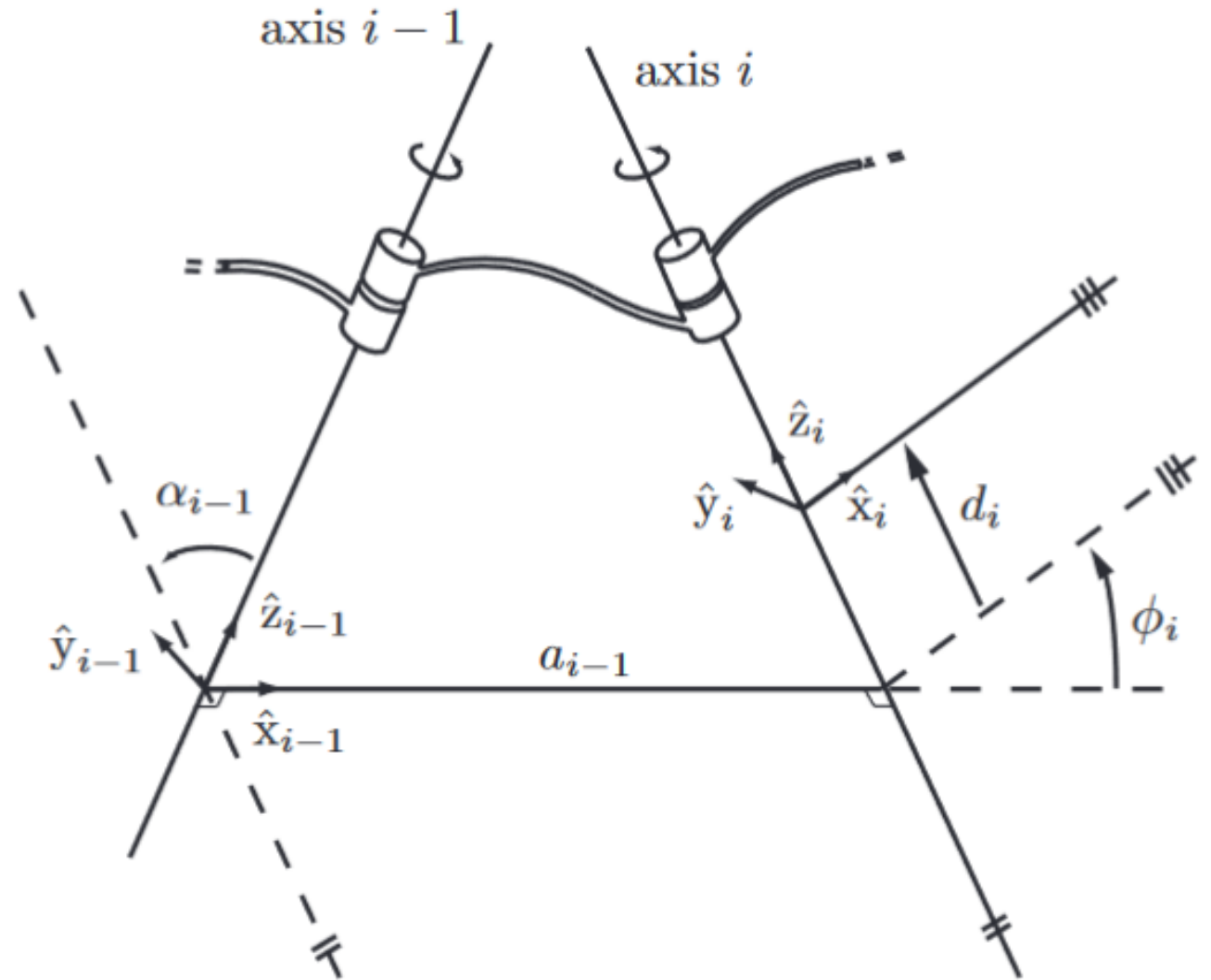
Denavit-Hartenberg Reference Frame Layout

Produced by Ethan Tira-Thompson



Rule 1

Assigning $\hat{z}_i$ axis coincide with joint i and $\hat{z}_{i-1}$ axis coincide with joint axis $i-1$. the direction of positive rotation is determined by the right hand rule.

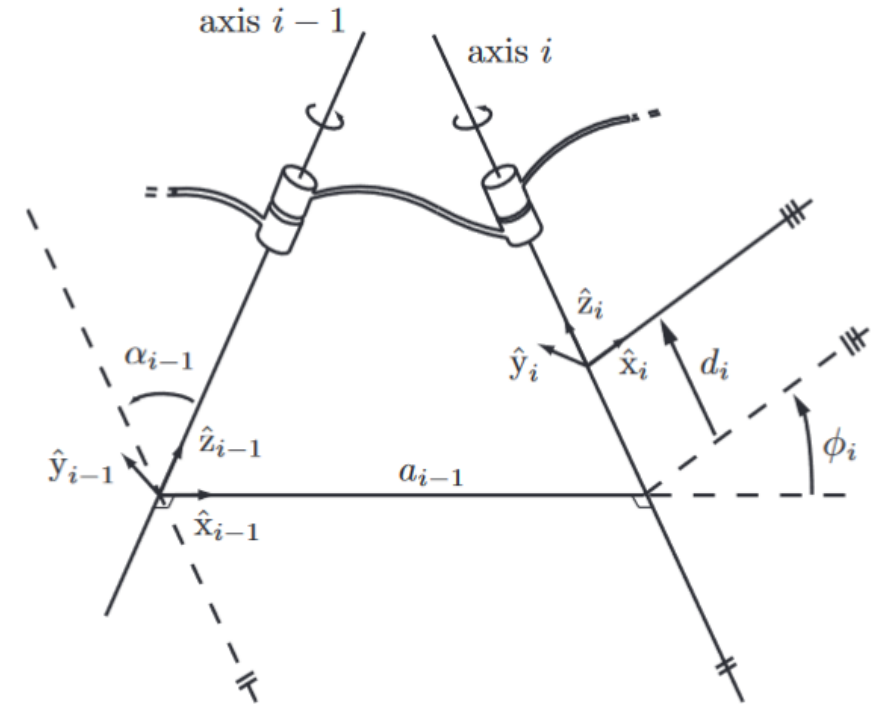


Rule 2: Origin of the link reference frame

Find the line segment that orthogonally intersects both the joint axes $\hat{z}_{i-1}$ and $\hat{z}_i$.

The $\hat{x}$ -axis is chosen to be in the direction of the mutually perpendicular line pointing from the (i-1)-axis to the i-axis.

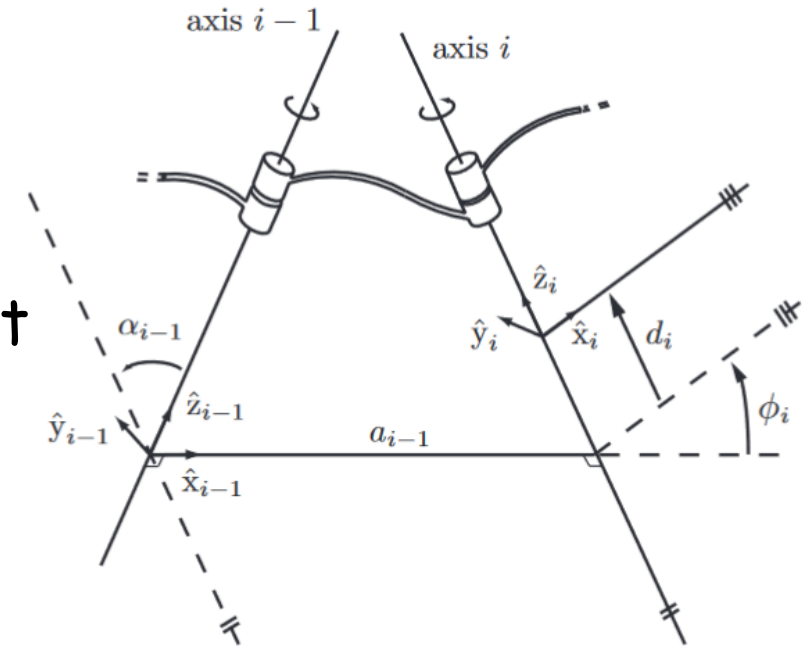
The $\hat{y}$ -axis is then uniquely determined from the cross product $\hat{x} \times \hat{y} = \hat{z}$.



Rule 3: The D-H parameters

Define four parameters that exactly specify $T_{i-1,i}$.

1. The length of the mutually perpendicular line, denoted by the scalar a_{i-1} is called the **link length** of link $i-1$. Despite its name, this link length does not necessarily correspond to the actual length of the physical link.
2. The **link twist** α_{i-1} between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$ (Rotation along z-axis).
3. The link offset d_i is the distance from the intersection of $\hat{x}_{i-1}$ and $\hat{z}_i$ to the origin of the link- i frame (displacement along z-axis)
4. The joint angle Φ_i is the angle from $\hat{x}_{i-1}$ to $\hat{x}_i$ about Z_{i-1} . (Rotation along x-azxs)



Manipulator Forward Kinematic

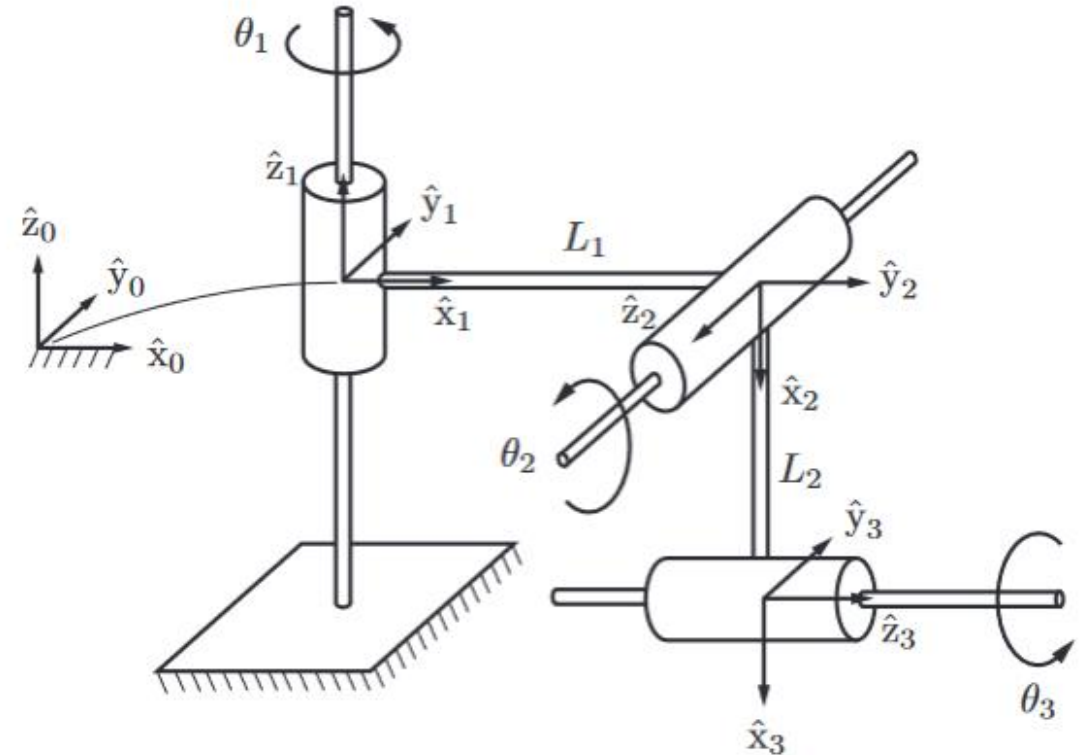
$$\begin{aligned}
 A_i &= R_{z,\theta_i} \text{Trans}_{z,d_i} \text{Trans}_{x,a_i} R_{x,\alpha_i} \quad (3.10) \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_ic_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_is_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Example: 3R Spatial open chain

i	α_{i-1}	a_{i-1}	d_i	ϕ_i
1	0	0	0	θ_1
2	90°	L_1	0	$\theta_2 - 90^\circ$
3	-90°	L_2	0	θ_3

D-H parameters

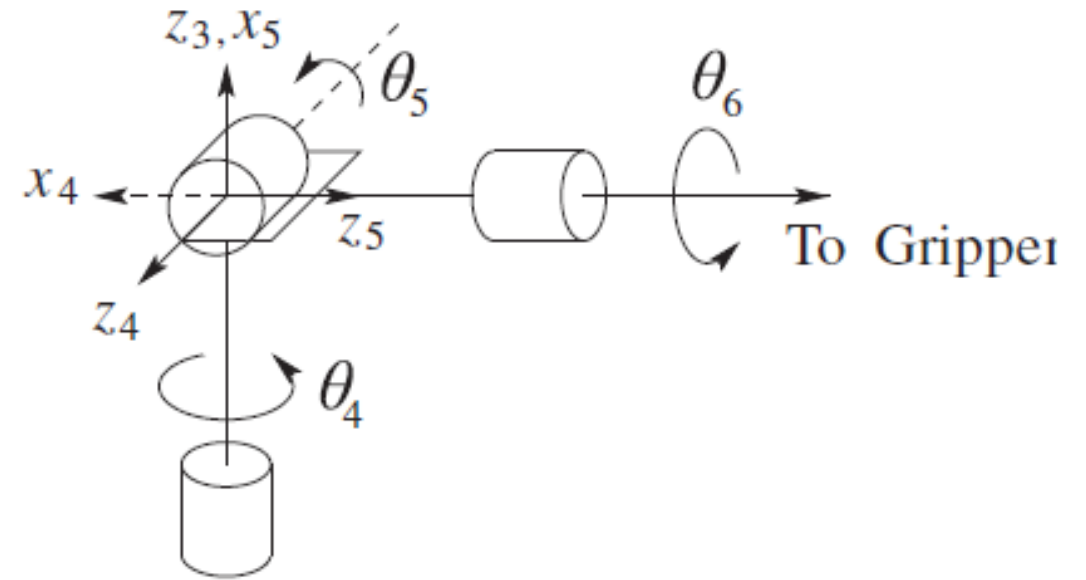
1. The length of the mutually perpendicular line, denoted by the scalar a_{i-1} is called the **link length** of link $i-1$. Despite its name, this link length does not necessarily correspond to the actual length of the physical link.
2. The **link twist** α_{i-1} between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$.
3. The link offset d_i is the distance from the intersection of $\hat{x}_{i-1}$ and $\hat{z}_i$ to the origin of the link- i frame.
4. The joint angle ϕ_i is the angle from $\hat{x}_{i-1}$ to $\hat{x}_i$.



Example: The Spherical Wrist

$$A_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link	a_i	α_i	d_i	θ_i
4	0	-90	0	θ_4
5	0	90	0	θ_5
6	0	0	d_6	θ_6



$$A_4 = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A_5 = \begin{bmatrix} c_5 & 0 & s_5 & 0 \\ s_5 & 0 & -c_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: The Spherical Wrist, cont.

$$\begin{aligned} T_6^3 &= A_4 A_5 A_6 \\ &= \begin{bmatrix} R_6^3 & o_6^3 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

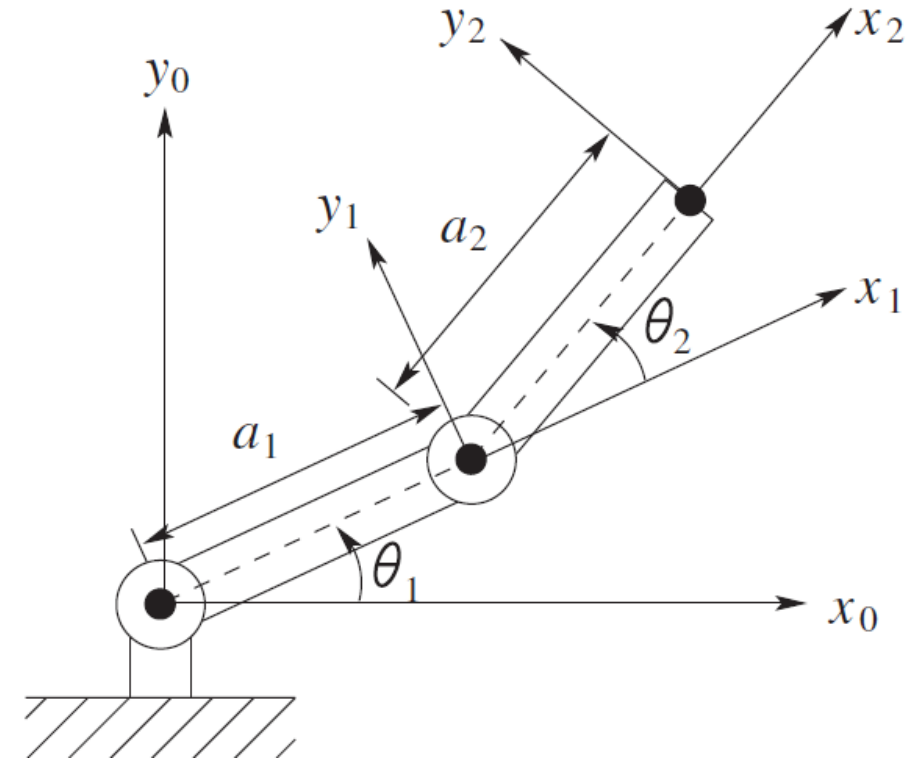
Example

DH parameters for 2-link planar manipulator:

Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2

$$T_1^0 = A_1$$

$$T_2^0 = A_1 A_2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



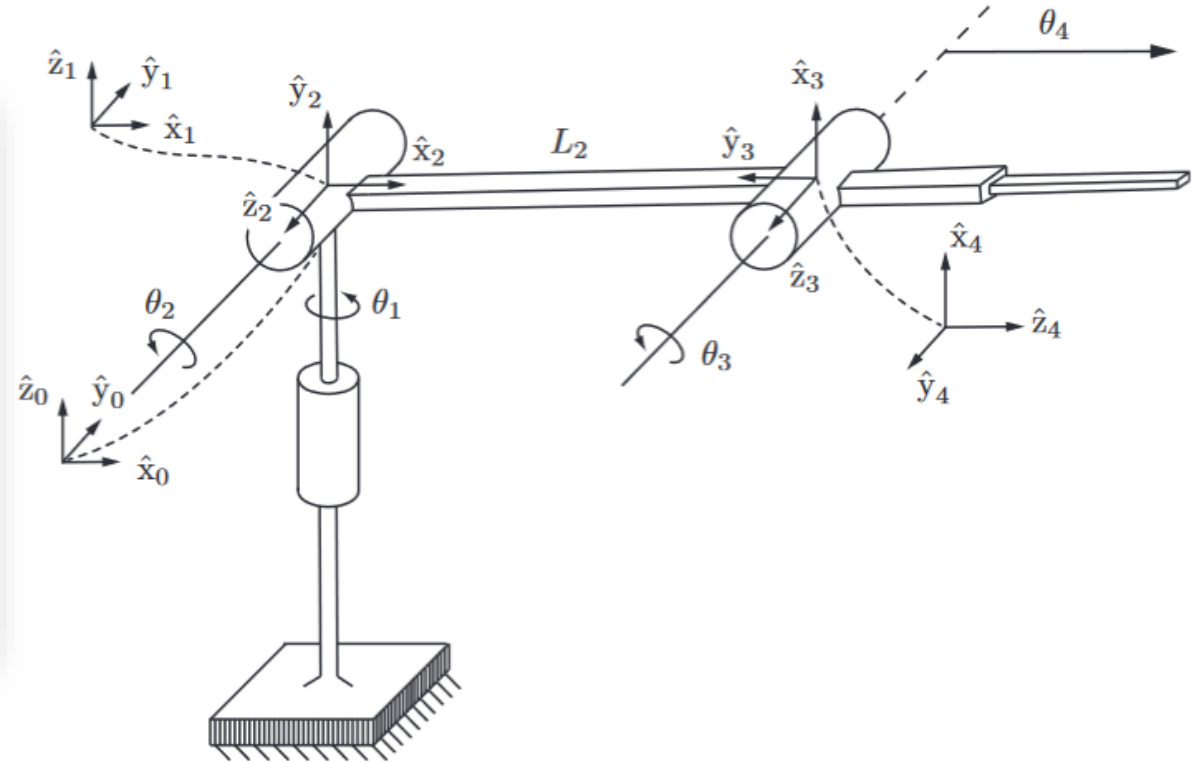
Notice that the first two entries of the last column of T_2^0 are the x and y components of the origin o_2 in the base frame; that is:

$$x = a_1 c_1 + a_2 c_{12}$$

$$y = a_1 s_1 + a_2 s_{12}$$

Example: RRRP spatial open chain

1. The length of the mutually perpendicular line, denoted by the scalar a_{i-1} is called the **link length** of link $i-1$. Despite its name, this link length does not necessarily correspond to the actual length of the physical link.
2. The **link twist** α_{i-1} between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$.
3. The link offset d_i is the distance from the intersection of $\hat{x}_{i-1}$ and $\hat{z}_i$ to the origin of the link- i frame.
4. The joint angle ϕ_i is the angle from $\hat{x}_{i-1}$ to $\hat{x}_i$.

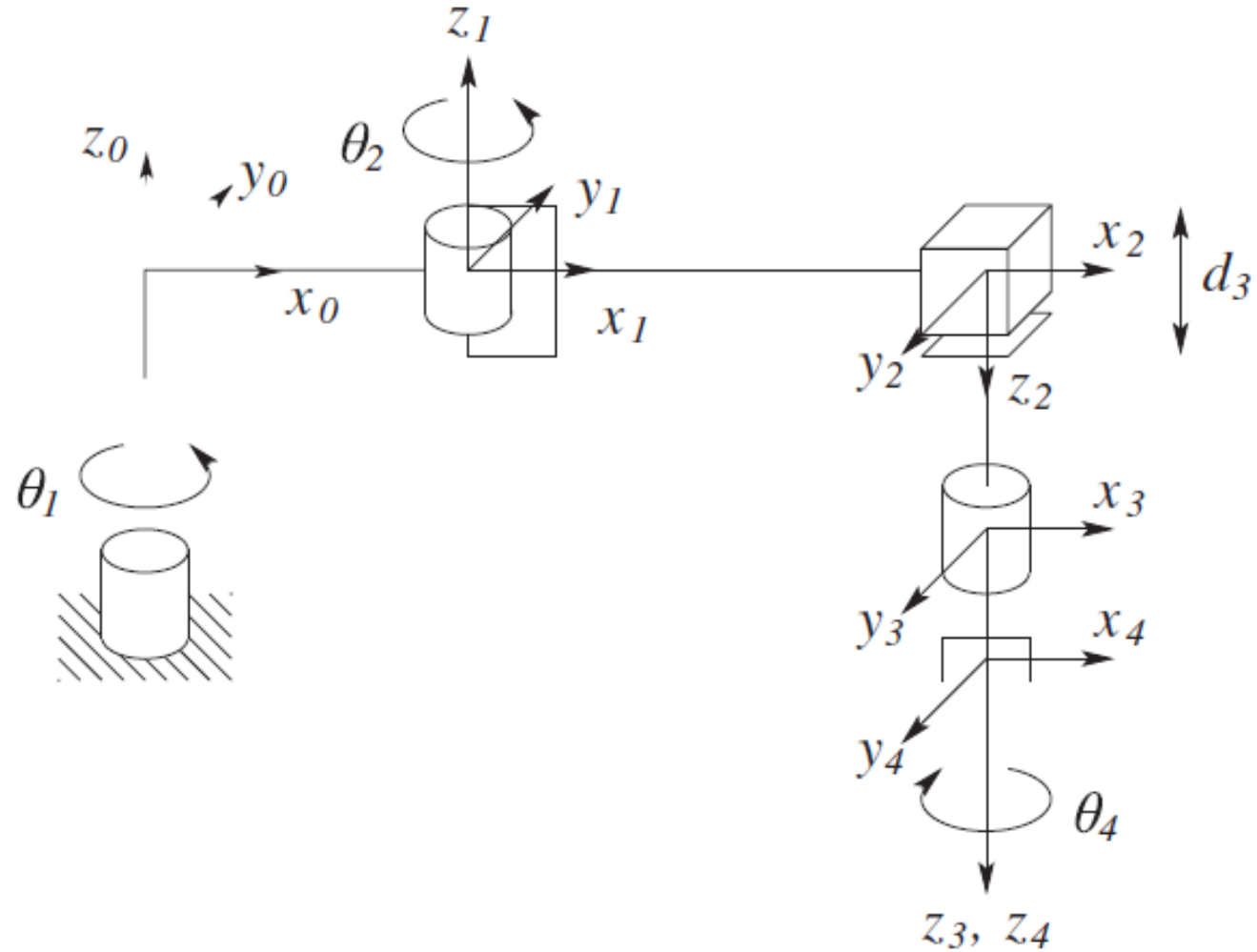


i	α_{i-1}	a_{i-1}	d_i	ϕ_i
1	0	0	0	θ_1
2	90°	0	0	θ_2
3	0	L_2	0	$\theta_3 + 90^\circ$
4	90°	0	θ_4	0

Assignment #1

SCARA manipulator

Determine DH parameters



Reading List:

Kevin M. Lynch and Frank C. Park, 2017, Modern Robotics, 1st Edition, Cambridge University Press, chapter 4.

Watching list

<https://www.youtube.com/watch?v=rA9tm0gTln8>

Questions?